4.

Right of , capacitors (Fig.) equal one

. Simplify to Fig. . First find .

Fig. Fig.

shut: final charge from to right of is

(positive, negative). Capacitor : ; plate

charge signs are as in Fig. .

Node charge balance: independent equations:

Loop voltages, e.g. , give

Both cap. and equivalent have voltage ; hence

Solving gives

From ,

During charging in Fig. , total circuit charge is ;

the source does work

Energy stored in the equivalent capacitor:

Energy conservation: heat dissipated in the resistor is